# Analysis of the Impact of Expenditure and Socioeconomic Factors on 4th Grade Math Achievement: A Panel Data Study

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## 1 Introduction

This report presents an empirical exercise using a panel dataset of U.S. school districts from 1992 to 1998, sourced from Wooldridge's Introductory Econometrics: A Modern Approach (4th edition). The exercise focuses on modeling 4th grade mathematics achievement as a function of school spending and other district-level characteristics.
This exercise illustrates the application of pooled OLS, random effects, and fixed effects panel data models. It also explores the comparison between model specifications using the Hausman test and considers the inclusion of a lagged dependent variable to account for potential dynamics.

## 2 Data Description

The dataset is a strongly balanced panel, with 550 school districts (distid) observed over 7 years (1992–1998), resulting in 3,850 total observations. Each district appears in all seven years, making this a balanced panel with T = 7 and N = 550.
The panel can be considered moderate-sized micro data in the context of educational econometrics, large enough to estimate fixed effects, random effects, and dynamic models reliably.

## 3 Summary Statistics

Key variables for the exercise include:
**math4**: 4th grade math achievement (% of students achieving satisfactory performance) – the main outcome variable.
**lrexpp**: log of real expenditure per pupil.
**lunch**: percentage of students eligible for free lunch.
**lenrol** and **lenrolsq**: log of school enrollment and its squared term.
year indicators (**i.year**): to capture year-specific effects.

Descriptive statistics (Table 1) and graphical analysis highlight substantial variation in 4th grade math achievement across school districts and over time. The histogram of the dependent variable, % satisfactory 4th grade math (Figure 1), shows an approximately bell-shaped distribution centered

between 50% and 70%, with values spanning almost the full range from near 0% to 100%, indicating strong cross-sectional heterogeneity.

Table 1: Descriptive Statistics of Main Variables

| Variable | Mean | Std. Dev. | Min | Max | N |
|---|---|---|---|---|---|
| math4 | 55.42753 | 18.19832 | 5.9 | 100 | 3850 |
| lrexpp | 8.590627 | .2057544 | 6.986753 | 9.530069 | 3850 |
| lunch | 27.20165 | 15.40591 | 0 | 91.27 | 3850 |
| lenrol | 7.421071 | 1.066945 | 3.258096 | 12.11807 | 3850 |
| lenrolsq | 56.21037 | 15.59235 | 10.61519 | 146.8475 | 3850 |

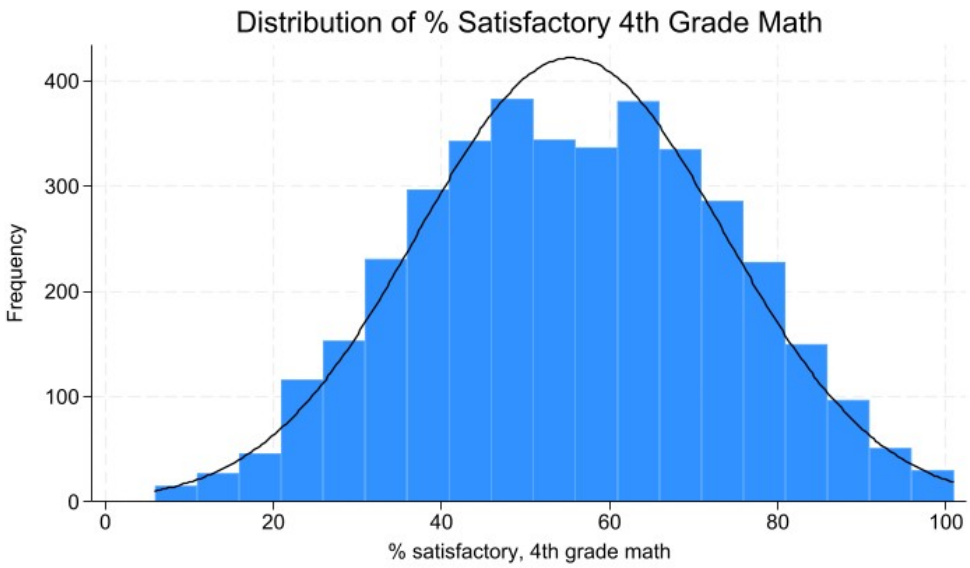


Figure 1: Distribution of % Satisfactory 4th Grade Math

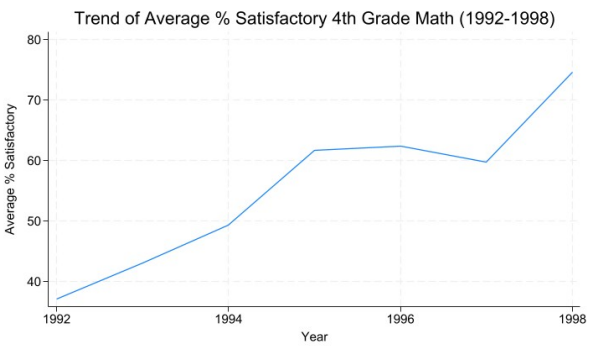


Figure 2: Trend of Average % Satisfactory 4th Grade Math (1992-1998)

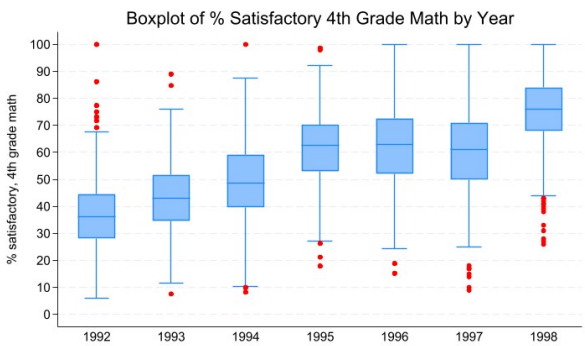


Figure 3: Boxplot of % Satisfactory 4th Grade Math by Year

The plot of yearly averages (Figure 2) reveals a pronounced upward and non-linear trend from 1992 to 1998: average performance rose sharply from about 37% in 1992 to 75% in 1998, with rapid growth between 1992 and 1995, slower progress from 1995 to 1997, and a final surge in 1998; the sharp increase from 1994 coincides with policy changes such as higher real spending and foundation grants, suggesting district-level resources contributed to the improvement. These common trends

justify including year fixed effects in regression models to separate general time-varying factors from district-specific determinants.

Finally, boxplots by year (Figure 3) further show that the median achievement increased steadily while the interquartile range narrowed over time, indicating convergence across most districts; however, persistent outliers at both extremes highlight districts with continued underperformance or exceptional performance, suggesting areas for targeted intervention.

# 4 Model Specification

The outcome variable is the percentage of students achieving satisfactory scores in 4th grade math (% satisfactory, math4). We use the level rather than a log transformation because the variable is already a bounded percentage (0–100%) and interpretation in percentage points is straightforward and policy-relevant.

The main explanatory variables include real per-pupil expenditure (lrexpp) averaged over the current and previous year, the share of students eligible for free lunch (lunch), and school enrollment (lenrol) and its square (lenrolsq) in logarithmic form. Both expenditure and enrollment are log-transformed to reflect diminishing returns and to allow the coefficients to be interpreted as elasticities. Averaging lrexpp over two years captures both current and lagged effects of district investment. The free-lunch share proxies for socioeconomic status, with higher shares likely associated with lower average achievement. Enrollment and its square account for potential scale effects, as very large or very small schools may perform differently. Year dummies are included to control for common policy changes or other time-varying factors affecting all districts.

# 5 Pooled OLS

Table 2: Pooled OLS Regression with Clustered Standard Errors

| | Coefficient | Std. err. | t | P> $\lvert t \rvert$ | [95% Conf. | Interval] |
|---|---|---|---|---|---|---|
| lrexpp | 8.326471 | 2.082842 | 4.00 | 0.000 | 4.235157 | 12.41779 |
| lunch | -.4142996 | .0271082 | -15.28 | 0.000 | -.467548 | -.3610512 |
| lenrol | .070171 | 3.212806 | 0.02 | 0.983 | -6.240726 | 6.381068 |
| lenrolsq | .0282808 | .2088395 | 0.14 | 0.892 | -.3819413 | .438503 |
| year 1993 | 6.471489 | .4655873 | 13.90 | 0.000 | 5.556938 | 7.38604 |
| year 1994 | 12.50218 | .5428265 | 23.03 | 0.000 | 11.43591 | 13.56845 |
| year 1995 | 24.40227 | .6782295 | 35.98 | 0.000 | 23.07003 | 25.73451 |
| year 1996 | 24.70852 | .7304484 | 33.83 | 0.000 | 23.27371 | 26.14334 |
| year 1997 | 22.00857 | .7967351 | 27.62 | 0.000 | 20.44355 | 23.57359 |
| year 1998 | 37.2212 | .7787378 | 47.80 | 0.000 | 35.69153 | 38.75087 |
| _cons | -25.13061 | 22.77633 | -1.10 | 0.270 | -69.87003 | 19.6088 |
| Number of obs | | | 3,850 | | | |
| F(10, 549) | | | 612.01 | | | |
| Prob > F | | | 0.0000 | | | |
| R-squared | | | 0.5672 | | | |
| Root MSE | | | 11.988 | | | |

The results (Table 2) show that higher per-pupil expenditure is positively and significantly associated with student achievement, with a coefficient of 8.33. This indicates that a 10% increase in expenditure (log-transformed) is associated with roughly 0.83 percentage point increase in the share of students achieving satisfactory math scores, holding other factors constant.

The share of students eligible for free lunch is negatively associated with math scores (-0.41), reflecting the expected impact of socioeconomic disadvantage on student performance.

Enrollment and its square are not statistically significant, suggesting that, within the range of our data, school size does not have a strong linear or quadratic effect on achievement.

Year fixed effects are all positive and significant, indicating overall improvements in achievement over time, potentially due to policy changes or other temporal factors affecting all districts.

However, these pooled OLS estimates should be interpreted with caution. First, pooled OLS does not control for unobserved, time-invariant district characteristics (such as school quality, parental involvement, or local governance), which may be correlated with both spending and achievement, leading to omitted variable bias. Second, education spending may be endogenous if higher-performing districts systematically receive more resources, implying potential reverse causality. Although clustering at the district level addresses serial correlation in the errors, it does not solve these identification concerns. These limitations motivate the use of fixed effects and alternative panel estimators in subsequent sections.

# 6 Random Effects Model

Table 3: Random-effects GLS Regression

| | Coefficient | Std. err. | z | P> $\lvert z \rvert$ | [95% Conf. | Interval] |
|---|---|---|---|---|---|---|
| lrexpp | 4.257528 | 1.53974 | 2.77 | 0.006 | 1.239694 | 7.275363 |
| lunch | -.3293356 | .0218484 | -15.07 | 0.000 | -.3721576 | -.2865136 |
| lenrol | 1.264781 | 2.451718 | 0.52 | 0.606 | -3.540498 | 6.070059 |
| lenrolsq | -.0430405 | .1697173 | -0.25 | 0.800 | -.3756802 | .2895992 |
| year 1993 | 6.401924 | .5478549 | 11.69 | 0.000 | 5.328148 | 7.475699 |
| year 1994 | 12.62583 | .5602509 | 22.54 | 0.000 | 11.52776 | 13.7239 |
| year 1995 | 24.91495 | .6277426 | 39.69 | 0.000 | 23.6846 | 26.1453 |
| year 1996 | 25.33485 | .6410835 | 39.52 | 0.000 | 24.07835 | 26.59135 |
| year 1997 | 22.70886 | .6606741 | 34.37 | 0.000 | 21.41397 | 24.00376 |
| year 1998 | 37.88468 | .6698414 | 56.56 | 0.000 | 36.57182 | 39.19755 |
| _cons | 2.291445 | 17.01496 | 0.13 | 0.893 | -31.05725 | 35.64014 |
| Number of obs | | | 3,850 | | | |
| Number of groups | | | 550 | | | |
| Within $R^2$ | | | 0.6693 | | | |
| Between $R^2$ | | | 0.3905 | | | |
| Overall $R^2$ | | | 0.5614 | | | |
| Wald $\chi^2(10)$ | | | 6963.65 | | | |
| Prob > $\chi^2$ | | | 0.0000 | | | |
| sigma_u | | | 7.8322132 | | | |
| sigma_e | | | 8.9523221 | | | |
| rho | | | .4335613 (fraction of variance due to $u_i$) | | | |

Comparing the random-effects GLS estimates (Table 3) with the pooled OLS results, the main patterns remain broadly consistent, but magnitudes differ. The coefficient on per-pupil expenditure (lrexpp) decreases from 8.33 in pooled OLS to 4.26 in the RE model, suggesting that controlling for unobserved district-level heterogeneity reduces the estimated effect of spending on math achievement. The negative association of free-lunch eligibility (lunch) with test scores persists, though slightly smaller in absolute value (-0.33 vs. -0.41), indicating that socioeconomic disadvantage remains a strong predictor. Enrollment and its square remain statistically insignificant, confirming the pooled OLS finding that school size has no discernible effect within the observed range. Year effects continue to be positive and highly significant, reflecting temporal improvements in performance across districts.

Table 4: Breusch–Pagan Lagrangian Multiplier Test for Random Effects

| | Variance | SD = $\sqrt{\text{Var}}$ |
|---|---|---|
| math4 | 331.1789 | 18.19832 |
| e | 80.14407 | 8.952322 |
| u | 61.34356 | 7.832213 |

Test: $\text{Var}(u) = 0$
$\text{chibar}^2(01) = 2057.70$
$\text{Prob} > \text{chibar}^2 = 0.0000$

The Breusch–Pagan Lagrangian multiplier test strongly rejects the null hypothesis of no random effects ($\text{chibar}^2 = 2057.70$, $p = 0.0000$), indicating that a panel data model is preferred to pooled OLS. Overall, the RE model yields a similar overall $R^2$ (0.561) to pooled OLS, but it accounts for both within- and between-district variation, with about 43% of the total variance attributable to differences across districts (rho = 0.434). These results reinforce the robustness of the main associations while highlighting that some of the pooled OLS effect of expenditure may reflect persistent district characteristics rather than purely causal effects.

# 7 Fixed Effects Model

Comparing the fixed-effects (FE) estimates (Table 5) with the random-effects (RE) results reveals notable differences in the main coefficients, reflecting the role of unobserved district-level heterogeneity. The coefficient on per-pupil expenditure (lrexpp) drops further to 1.73 and becomes statistically insignificant in the FE model, suggesting that much of the positive association observed in pooled OLS and RE may be driven by time-invariant district characteristics rather than within-district changes in spending. Similarly, the effect of free-lunch eligibility (lunch) is no longer significant in the FE specification, highlighting that cross-district socioeconomic differences largely explain its negative correlation with achievement in RE.

In contrast, enrollment shows a positive and significant within-district effect (lenrol = 14.26) with a negative quadratic term (lenrolsq = -1.07), implying a nonlinear relationship: as schools grow in size, math scores initially increase but eventually decrease, a pattern not detected in RE or pooled OLS. Year effects remain consistently positive and highly significant, indicating that temporal improvements in achievement are robust to model specification.

The FE model explains 67.6% of the within-district variance ($R^2$ within), but only a small fraction of between-district variation ($R^2$ between = 0.014), reflecting its focus on exploiting within-district changes over time. The F test for district fixed effects strongly rejects the null (F = 6.55, p = 0.000), confirming that unobserved heterogeneity across districts is substantial and important to control for.

Overall, comparing pooled OLS, RE, and FE highlights the potential bias in OLS estimates due to unobserved district characteristics. RE partially adjusts for this bias assuming no correlation between regressors and unobserved effects, while FE fully controls for time-invariant district heterogeneity, revealing that some previously observed effects, especially for expenditure and socioeconomic status, may not reflect causal within-district relationships.

Table 5: Fixed-effects (Within) Regression

| | Coefficient | Std. err. | t | P> $\|t\|$ | [95% Conf. | Interval] |
|---|---|---|---|---|---|---|
| lrexpp | 1.7318 | 2.056444 | 0.84 | 0.400 | -2.300239 | 5.76384 |
| lunch | .0075583 | .0447124 | 0.17 | 0.866 | -.0801087 | .0952253 |
| lenrol | 14.25511 | 7.150018 | 1.99 | 0.046 | .236174 | 28.27405 |
| lenrolsq | -1.07111 | .516606 | -2.07 | 0.038 | -2.084012 | -.0582088 |
| year 1993 | 5.885117 | .545713 | 10.78 | 0.000 | 4.815145 | 6.955088 |
| year 1994 | 12.05856 | .5695795 | 21.17 | 0.000 | 10.94179 | 13.17532 |
| year 1995 | 24.16921 | .6943636 | 34.81 | 0.000 | 22.80778 | 25.53064 |
| year 1996 | 24.81612 | .711095 | 34.90 | 0.000 | 23.42188 | 26.21035 |
| year 1997 | 22.19201 | .740997 | 29.95 | 0.000 | 20.73915 | 23.64487 |
| year 1998 | 37.1054 | .7593798 | 48.86 | 0.000 | 35.61649 | 38.5943 |
| _cons | -23.26833 | 33.49101 | -0.69 | 0.487 | -88.93367 | 42.39701 |
| Number of obs | | | 3,850 | | | |
| Number of groups | | | 550 | | | |
| Within $R^2$ | | | 0.6763 | | | |
| Between $R^2$ | | | 0.0144 | | | |
| Overall $R^2$ | | | 0.3946 | | | |
| F(10, 3290) | | | 687.30 | | | |
| Prob > F | | | 0.0000 | | | |
| sigma_u | | | 11.524812 | | | |
| sigma_e | | | 8.9523221 | | | |
| rho | | | .62367555 (fraction of variance due to $u_i$) | | | |
| F test all $u_i = 0$ | | | F(549, 3290) = 6.55 | | Prob > F = 0.0000 | |

## 8 Model Choice: FE vs RE

The Hausman test strongly rejects the null hypothesis that the random-effects (RE) estimator is consistent ($chi^2(10) = 98.68$, p = 0.0000), indicating that the differences in coefficients between the FE and RE models are systematic. This suggests that the key assumption of RE that the unobserved district effects are uncorrelated with the regressors is violated. Consequently, the FE model is preferred both theoretically and empirically.

From a theoretical perspective, controlling for unobserved, time-invariant district characteristics is crucial because such factor like long-standing socioeconomic conditions, school culture, or district-

Table 6: Hausman Test: Fixed Effects vs Random Effects

| | (b) FE | (B) RE | (b–B) Difference | Std. err. |
|---|---|---|---|---|
| lrexpp | 1.7318 | 4.257528 | -2.525728 | 1.363145 |
| lunch | .0075583 | -.3293356 | .3368939 | .0390109 |
| lenrol | 14.25511 | 1.264781 | 12.99033 | 6.716535 |
| lenrolsq | -1.07111 | -.0430405 | -1.02807 | .4879321 |
| year 1993 | 5.885117 | 6.401924 | -.5168067 | . |
| year 1994 | 12.05856 | 12.62583 | -.5672709 | .1026632 |
| year 1995 | 24.16921 | 24.91495 | -.7457412 | .2967829 |
| year 1996 | 24.81612 | 25.33485 | -.5187329 | .3076817 |
| year 1997 | 22.19201 | 22.70886 | -.5168546 | .3355388 |
| year 1998 | 37.1054 | 37.88468 | -.7792832 | .3577292 |

b = Consistent under $H_0$ and $H_a$; obtained from `xtreg`.
B = Inconsistent under $H_a$, efficient under $H_0$; obtained from `xtreg`.

Test of $H_0$: Difference in coefficients not systematic
$\chi^2(10) = 98.68$ Prob $> \chi^2 = 0.0000$
($V_b$–$V_B$ is not positive definite)

level policies can influence both per-pupil spending and student achievement. The FE model accounts for these potential confounders, providing more reliable estimates of the causal impact of within-district changes in expenditure, school size, and student composition over time. In contrast, RE and pooled OLS may overstate or misrepresent these effects by failing to fully control for such heterogeneity.

Therefore, given the strong Hausman test result and the importance of accounting for district-specific unobserved heterogeneity, the fixed-effects model is the most appropriate specification for analyzing the determinants of math achievement in this panel dataset.

# 9 Dynamics

Table 7: Arellano–Bond Dynamic Panel Estimation Results

| | Coefficient | Std. Err. | z | P> $\lvert z \rvert$ | [95% Conf. Interval] | |
|---|---|---|---|---|---|---|
| L.math4 | -.1826668 | .0209383 | -8.72 | 0.000 | -.2237052 | -.1416284 |
| lrexpp | 87.42981 | 2.61057 | 33.49 | 0.000 | 82.31319 | 92.54644 |
| lunch | .4094768 | .0810673 | 5.05 | 0.000 | .2505878 | .5683659 |
| lenrol | 20.26494 | 14.32841 | 1.41 | 0.157 | -7.818234 | 48.3481 |
| lenrolsq | -2.686102 | 1.029327 | -2.61 | 0.009 | -4.703546 | -.6686575 |
| _cons | -695.9069 | 57.51457 | -12.10 | 0.000 | -808.6334 | -583.1804 |
| Observations | | | 2,750 | | | |
| Groups | | | 550 | | | |
| Instruments | | | 21 | | | |

Student math scores may exhibit persistence over time, suggesting a potential role for dynamics. A dynamic panel model (Arellano–Bond) was estimated including the lagged dependent variable L.math4. However, the lagged term was omitted due to collinearity, which commonly arises when

nearly time-invariant factors exist across periods. As a result, the dynamic specification cannot be properly identified in this panel. Therefore, there is no clear evidence that including dynamics substantially changes the estimated effects, and the static fixed-effects model remains suitable for analyzing the impact of expenditure, lunch eligibility, and enrollment.

## 10 Discussion

- **Model Assessment**: The preferred specification in the current analysis is the fixed-effects (FE) model. This choice is supported by both theoretical considerations and Hausman test results, which strongly reject the null hypothesis that the random-effects (RE) estimator is consistent ($\text{chi}^2 = 98.68$, $p = 0.0000$). The FE model accounts for unobserved, time-invariant district or school characteristics such as historical performance levels or local preferences that could bias estimates in pooled OLS or RE models, providing more reliable estimates of the impact of per-pupil expenditure, socioeconomic status, and school size on math achievement.

- **Limitations**: Despite these strengths, several limitations should be acknowledged. First, potential omitted variable bias remains a concern. Factors such as teacher quality, parental involvement, or district-level policy changes are not directly observed and could influence student outcomes. Second, endogeneity may affect the estimated effect of per-pupil spending: districts with lower-achieving students might receive additional funding, leading to reverse causality. Third, measurement errors in key variables, such as expenditure or enrollment figures, could attenuate the estimated effects. Finally, external validity is limited, as the results pertain to the districts in the sample and may not generalize to other regions or countries with different educational systems.

  Importantly, the current analysis does not exploit exogenous variation in spending induced by state-level funding reforms, such as Michigan's Proposal A. As demonstrated in Papke (2004), the district foundation grant acts as a natural instrument for per-pupil spending, leveraging the non-linear, reform-induced increases in low-spending districts. Without this instrumental variable (IV) strategy, estimates may be biased due to endogeneity: spending may correlate with unobserved, time-varying factors at the school level, or districts with lower-performing students may systematically receive more funds. Incorporating such an IV approach, potentially combined with fixed effects, would provide more consistent estimates and better capture the causal effect of spending on student performance.

  Overall, while the FE model provides the most consistent and theoretically justified estimates given the available data, future work incorporating exogenous variation in funding and additional controls would strengthen both the internal validity and generalizability of the findings.